

5G RAN

Transmission Resource Management Feature Parameter Description

Issue	Draft A
Date	2025-12-31



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....1

1.1 5G RAN10.1 Draft A (2025-12-31).....1

2 About This Document.....3

2.1 General Statements.....3

2.2 Features in This Document.....3

2.3 Differences Between NSA and SA.....4

2.4 Differences Between High Frequency Bands and Low Frequency Bands.....4

3 Overview.....5

4 Transmission Resource Configuration and Mapping.....9

4.1 Principles.....9

4.1.1 Transmission Resource Configuration.....9

4.1.1.1 Base Station Transmission Resource Configuration.....9

4.1.1.1.1 Physical Port.....10

4.1.1.1.2 Transmission Resource Group.....10

4.1.1.1.3 Endpoint.....12

4.1.2 Transmission Resource Mapping.....13

4.1.2.1 Base Station Transmission Resource Mapping.....13

4.1.2.1.1 QoS Objectives.....13

4.1.2.1.2 Mapping Between Service Types and DSCPs.....16

4.1.2.1.3 Mapping Between DSCPs and VLAN Priorities.....19

4.2 Network Analysis.....20

4.2.1 When to Use.....20

4.2.2 Required Information.....21

4.2.3 Benefits.....21

4.2.4 Impacts.....21

4.3 Requirements.....21

4.3.1 Licenses.....21

4.3.2 Functions.....21

4.3.2.1 Prerequisite Functions.....21

4.3.2.2 Mutually Exclusive Functions.....21

4.3.2.3 Function Impacts.....21

4.3.3 Hardware.....22

4.3.4 Others..... 22

4.4 Operation and Maintenance..... 22

4.4.1 Data Configuration..... 22

4.4.1.1 Data Preparation..... 22

4.4.1.2 Using MML Commands..... 29

4.4.1.3 Using the MAE-Deployment..... 30

4.4.2 Activation Verification..... 31

4.4.3 Network Monitoring..... 32

5 Transmission Congestion Control..... 33

5.1 Principles..... 33

5.1.1 Base Station Transmission Congestion Control..... 33

5.1.1.1 Traffic Shaping..... 34

5.1.1.2 Queue Scheduling..... 37

5.1.1.3 Backpressure Algorithm..... 39

5.2 Network Analysis..... 41

5.2.1 When to Use..... 41

5.2.2 Benefits..... 42

5.2.3 Impacts..... 42

5.3 Requirements..... 42

5.3.1 Licenses..... 42

5.3.2 Functions..... 42

5.3.2.1 Prerequisite Functions..... 42

5.3.2.2 Mutually Exclusive Functions..... 42

5.3.2.3 Function Impacts..... 42

5.3.3 Hardware..... 42

5.3.4 Others..... 43

5.4 Operation and Maintenance..... 43

5.4.1 Data Configuration..... 43

5.4.1.1 Data Preparation..... 43

5.4.1.2 Using MML Commands..... 47

5.4.1.3 Using the MAE-Deployment..... 49

5.4.2 Activation Verification..... 49

5.4.3 Network Monitoring..... 50

6 Parameters..... 51

7 Counters..... 53

8 Glossary..... 54

9 Reference Documents..... 55

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for X2 interfaces in SA networking. For details, see: • 3 Overview • 4.1.2.1.1 QoS Objectives • 4.1.2.1.2 Mapping Between Service Types and DSCPs • 4.2.1 When to Use • 4.4.1.2 Using MML Commands • 5.1.1 Base Station Transmission Congestion Control • 5.1.1.3 Backpressure Algorithm	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FBFD-010018	Transmission QoS Management	4 Transmission Resource Configuration and Mapping
		5 Transmission Congestion Control

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Transmission Resource Configuration and Mapping	None	4 Transmission Resource Configuration and Mapping
Transmission Congestion Control	None	5 Transmission Congestion Control

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Transmission Resource Configuration and Mapping	None	4 Transmission Resource Configuration and Mapping
Transmission Congestion Control	None	5 Transmission Congestion Control

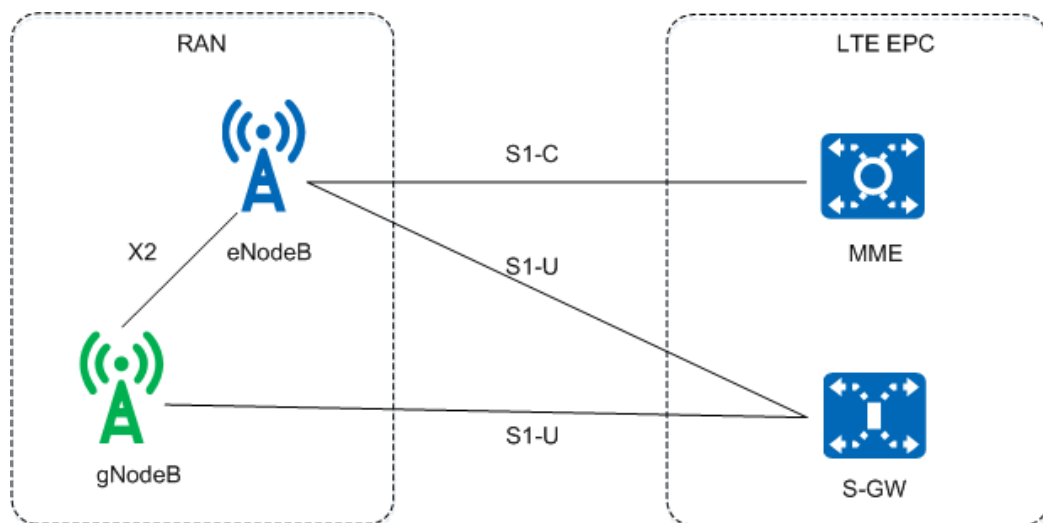
3 Overview

Transmission resources are one type of resources on the access network and refer to the transmission bandwidth resources of logical interfaces S1 and X2 in the NR NSA architecture and logical interfaces NG, Xn, eXn, and SA X2 in the NR SA architecture. Transmission resource management involves transmission resource configuration and mapping and transmission congestion control.

- Transmission resource configuration and mapping
 - Transmission resource configuration: Physical ports, transmission resource groups, and endpoints are configured to implement accurate and precise bandwidth management. For details, see [4.1.1 Transmission Resource Configuration](#).
 - Mapping of services to transmission resources: Service types are mapped to differentiated services code points (DSCPs), and then to virtual local area network (VLAN) priorities based on quality of service (QoS) requirements. This helps differentiate services and ensure that transmission resources are allocated fairly. For details, see [4.1.2 Transmission Resource Mapping](#).
- Transmission congestion control: Congestion control ensures the QoS of high-priority services by using differentiated flow control based on service types when a transmission network is congested. For details, see [5 Transmission Congestion Control](#).

[Figure 3-1](#) shows the logical architecture of the S1 and X2 interfaces for base stations in an integrated NSA deployment.

Figure 3-1 Logical interface architecture for base stations in an integrated NSA deployment



Interfaces on the gNodeB side in the NSA architecture are as follows:

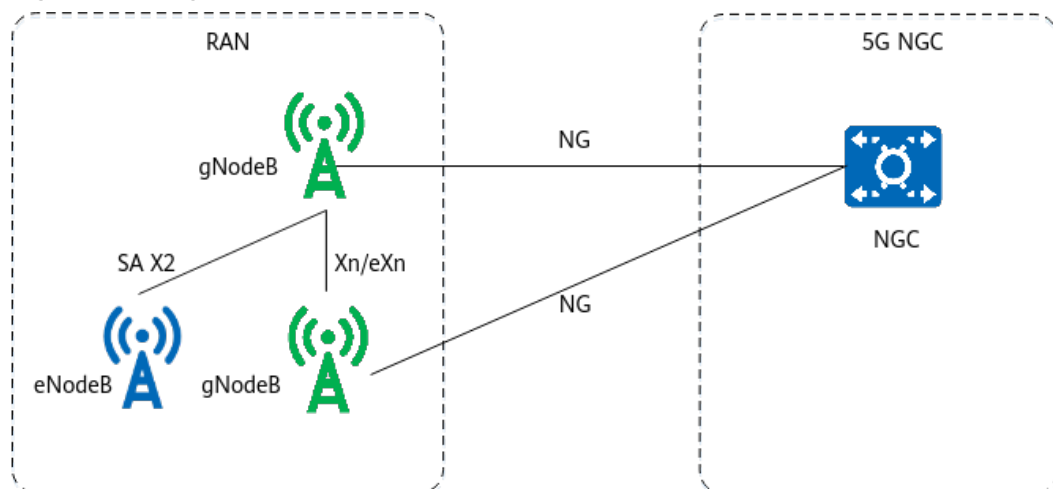
- The S1-U interface is used by a gNodeB to exchange user-plane data with an LTE serving gateway (S-GW) in the NSA architecture.
- The X2 interface is a logical interface between a gNodeB and an eNodeB in the NSA architecture. It exchanges control-plane and user-plane information between the gNodeB and the eNodeB.

NOTE

This document only describes interfaces on the gNodeB side. For interfaces on the eNodeB side in the NSA architecture, see *Transmission Resource Management in eRAN Feature Documentation*.

Figure 3-2 shows the logical architecture of the NG, Xn, eXn, and SA X2 interfaces for base stations in the SA architecture.

Figure 3-2 Logical interface architecture for base stations in the SA architecture



Interfaces on the gNodeB side in the SA architecture are as follows:

- The NG interface is used to transmit user-plane and control-plane data between a gNodeB and a 5G NGC in the SA architecture.
- The Xn interface is a logical interface between gNodeBs in the SA architecture. It exchanges control-plane and user-plane information between gNodeBs.
- The eXn interface is a logical interface between gNodeBs in the SA architecture. It carries coordination data between gNodeBs.
- The SA X2 interface is a logical interface between a gNodeB and an eNodeB in the SA architecture. It exchanges control-plane and user-plane information between the gNodeB and the eNodeB. For details about the SA X2 interface, see *X2 Self-Management in SA Networking*.

Transmission configuration models include the new and old models. For details, see *IPv4 Transmission*.

Table 3-1 lists the QoS requirements of the S1/NG interface on the transmission network.

Table 3-1 QoS requirements of the S1/NG interface on the transmission network

QoS Indicator	Recommended Value
One-way delay (ms) (integrated deployment)	5
Packet loss rate	1.0×10^{-6}

NOTE

A recommended value is one that meets QoS requirements for most services except for low-delay and high-bandwidth services such as strong interactive AR. QoS requirements of services requiring a low delay or a high bandwidth are met by deploying service servers close to the RAN side.

Table 3-2 lists the QoS requirements of the X2/Xn/eXn/SA X2 interface on the transmission network.

Table 3-2 QoS requirements of the X2/Xn/eXn/SA X2 interface on the transmission network

Interface Type	QoS Indicator	Recommended Value
Xn	One-way delay (ms) (without inter-base-station dual connectivity)	10
Xn	Packet loss rate	1.0×10^{-6}
X2	One-way delay (ms) (NSA Option 3x inter-base-station dual connectivity)	5
X2	Packet loss rate	1.0×10^{-6}

Interface Type	QoS Indicator	Recommended Value
eXn	One-way delay (ms) (relaxed backhaul networking)	4
eXn	Packet loss rate	1.0×10^{-6}
SA X2	One-way delay (ms)	5
SA X2	Packet loss rate	1.0×10^{-6}

4 Transmission Resource Configuration and Mapping

4.1 Principles

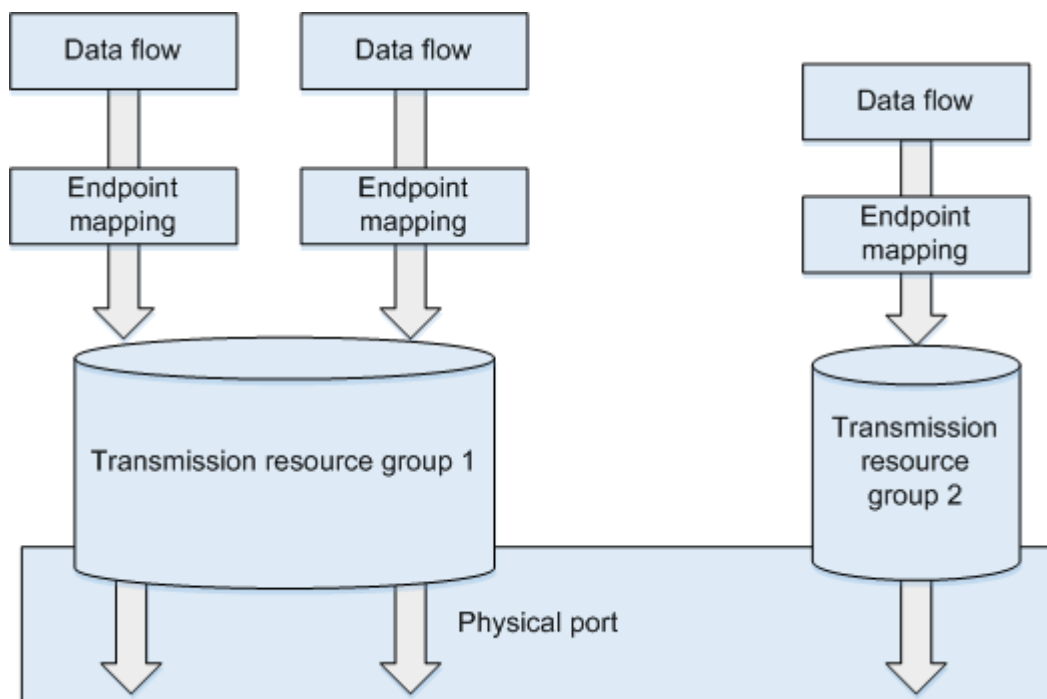
Transmission resource configuration and mapping is the basis of transmission resource management. It includes the configuration of transmission resources and mapping of services to these resources.

4.1.1 Transmission Resource Configuration

4.1.1.1 Base Station Transmission Resource Configuration

Base station transmission resources include physical ports, transmission resource groups, and endpoints, which are used to manage incoming data flows, as shown in [Figure 4-1](#).

Figure 4-1 Transmission resource management



4.1.1.1.1 Physical Port

Physical ports of a base station include FE/GE/10GE ports. For details about the physical ports on each board, see *Hardware Description* in *3900 & 5900 Series Base Station Product Documentation*.

4.1.1.1.2 Transmission Resource Group

Using transmission resource groups enables separate management of bandwidths of different data flows over the same physical port. Transmission resource groups allow for dynamic bandwidth sharing between data flows in cascading scenarios and ensure fair bandwidth allocation.

There are two types of transmission resource groups.

- **Default transmission resource group**
Each physical port has a default or adaptive port transmission resource group, which does not need to be created. User-configured attributes of a default port transmission resource group can be modified, but those of an adaptive port transmission resource group cannot. Users can modify the attributes of a default port transmission resource group only after the group has been added.
- **Dedicated transmission resource group**
This type of transmission resource group must be created by users. Each dedicated transmission resource group corresponds to only one physical port, which is specified by the **RSCGRP.PN** (old model)/**IPRSCGRP.PORTID** (new model) parameter.

A physical port schedules the default transmission resource group, which is automatically generated by the base station, based on the weight of the transmission resource group. The weight is specified by the **RSCGRP.TXCIR** (old

model)/**IPRSCGRP.TXCIR** (new model) and **RSCGRP.TXBW** (old model)/**IPRSCGRP.TXBW** (new model) parameters. The default values of these parameters indicate the maximum bandwidth capabilities supported by the physical port. The maximum bandwidth capabilities supported by physical ports depend on the boards. Therefore, the weight of each transmission resource group must be adjusted based on a bandwidth plan. Otherwise, the effective bandwidth of a transmission resource group may be inconsistent with the expected bandwidth.

Transmission resource groups have single-rate and dual-rate modes. Users can select a rate mode by setting the **GTRANSPARA.RATECFGTYPE** parameter. Different rate modes have different data configuration requirements. In addition, the methods of scheduling transmission resource groups of physical ports also vary.

NOTE

In this document, when **GTRANSPARA.TRANSCFGMODE** is set to **OLD**, the old model is used; when **GTRANSPARA.TRANSCFGMODE** is set to **NEW**, the new model is used.

Single-Rate Mode

In this mode, only one bandwidth can be configured for the transmit direction, which is the peak transmit bandwidth specified by **RSCGRP.TXBW** (old model)/**IPRSCGRP.TXBW** (new model). The total throughput of all services in a transmission resource group cannot exceed the peak bandwidth of this group.

In single-rate mode, transmission resource groups of physical ports are scheduled as follows:

- When **GTRANSPARA.LPSCHSW** is set to **ENABLE(Enable)**, the weighted round robin (WRR)-based weight specified by **RSCGRP.WEIGHT** (old model)/**IPRSCGRP.WEIGHT** (new model) is used for scheduling transmission resource groups.
- When **GTRANSPARA.LPSCHSW** is set to **DISABLE(Disable)**, the WRR-based weight specified by **RSCGRP.TXBW** (old model)/**IPRSCGRP.TXBW** (new model) is used for scheduling transmission resource groups.

Dual-Rate Mode

In this mode, two bandwidths can be configured for the transmit direction: the committed bandwidth and the peak bandwidth. The bandwidth of all services in a transmission resource group cannot be less than the committed bandwidth and cannot exceed the peak bandwidth. When multiple transmission resource groups share the same physical port and the physical port becomes congested, dual-rate mode better ensures committed bandwidth for each transmission resource group, compared with the single-rate mode.

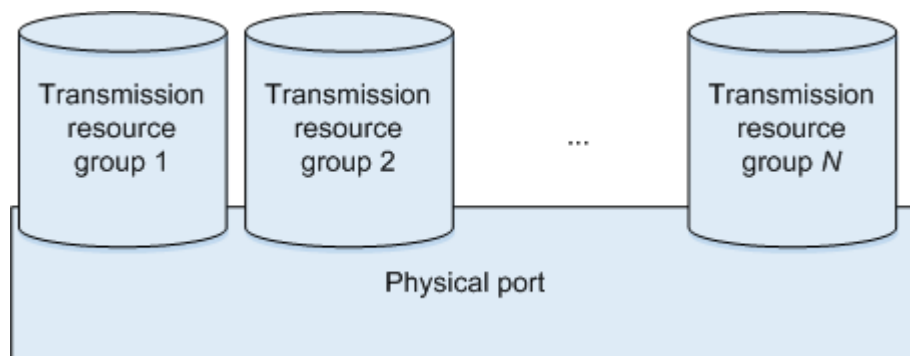
The transmit committed information rate (CIR) is configured using the **RSCGRP.TXCIR** (old model)/**IPRSCGRP.TXCIR** (new model) parameter, and the transmit peak information rate (PIR) is configured using the **RSCGRP.TXPIR** (old model)/**IPRSCGRP.TXPIR** (new model) parameter.

Figure 4-2 shows transmission resource groups configured over a physical port. A transmission resource group can map to one physical port, but a single physical

port may be mapped to multiple transmission resource groups. The configuration requirements are as follows:

- The transmit CIR of one transmission resource group must be less than the transmit PIR.
- The sum of committed bandwidths of all transmission resource groups is less than or equal to the transmit bandwidth of the physical port.

Figure 4-2 Transmission resource groups configured over a physical port



NOTE

Transmit bandwidth of a physical port = $\text{Min}(\text{Actual data sending rate of the physical port}, \text{LR.CIR (old model)}/\text{PORTLR.CIR (new model)})$

In dual-rate mode, transmission resource groups of physical ports are scheduled as follows:

1. WRR-based scheduling is first used for transmission resource groups where transmit CIR is not satisfied. The scheduling weight of each transmission resource group is specified by **RSCGRP.TXCIR** (old model)/**IPRSCGRP.TXCIR** (new model).
2. WRR-based scheduling is then used for transmission resource groups where transmit CIR is satisfied. The scheduling weight of each transmission resource group is specified by **RSCGRP.WEIGHT** (old model)/**IPRSCGRP.WEIGHT** (new model).

The base station uses the dual-rate mode to better match the actual transmission bandwidth. This ensures that the bandwidth used by non-flow-controllable services does not exceed the CIR, thereby preventing packet loss on the transport network and improving the bandwidth usage.

4.1.1.1.3 Endpoint

Data flows for each type of services must be mapped to transmission resource groups that have been configured. NR only supports the endpoint configuration mode. In this mode, the control- and user-plane links are set up automatically based on the source port details and the destination port details carried in the signaling. Endpoints are configured to map services to transmission resource groups. For details about the endpoint configuration mode, see *X2 and S1 Self-Management in NSA Networking*.

4.1.2 Transmission Resource Mapping

4.1.2.1 Base Station Transmission Resource Mapping

Data flows must be mapped to transmission resource groups based on the following rules:

- Local user-plane data uses either the default or a dedicated transmission resource group.
In endpoint configuration mode, the **ADD EP2RSCGRP** (old model)/**ADD EP2IPRSCGRP** (new model) command is used to specify a transmission resource group for local user-plane data.
- Control-plane, OM, IP clock, and passing-by data use the default transmission resource group by default. Alternatively, the **ADD IP2RSCGRP** (old model for IPv4 transmission)/**ADD IP2IPRSCGRP** (new model) command can be used to specify dedicated groups for these data flows based on destination IP addresses.

4.1.2.1.1 QoS Objectives

A base station transmits the S1/X2/NG/Xn/eXn/SA X2 interface data, OM data, IP clock data, and co-transmission data. QoS requirements vary with the service type. By configuring different priorities for these services, services can be differentiated based on their QoS.

QoS Requirements

Table 4-1 lists QoS requirements for data transmitted by the base stations.

Table 4-1 QoS requirements

Service Type	QoS Requirement	Description
User-plane real-time services: - Guaranteed bit rate (GBR) services, with a QoS class identifier (QCI)/5G QoS identifier (5QI) of 1 to 4, 65 to 67, 71 to 76, or 82 to 86 - Non-flow-controllable services in non-GBR services, with a QCI/5QI of 5 or 69 by default	The required bandwidths must be guaranteed.	The packet loss rate must be controlled. Increased delay due to a large amount of buffer data must be avoided. Otherwise, service quality is severely affected. When a gNodeB does not have enough bandwidth available in the uplink, the available bandwidth must be preferentially allocated to real-time services.

Service Type	QoS Requirement	Description
User-plane non-real-time services: Flow-controllable services in non-GBR services, with a QCI/5QI of 6 to 9, 70, 79, or 80 by default	The priority of this type of services is lower than that of real-time services.	When there is not enough bandwidth, service throughput can be decreased and data can be buffered to guarantee the basic quality of non-real-time services.
Control-plane services	The priority of this type of services is high.	Control-plane signaling has small traffic volume but is closely related to network KPIs. Bandwidth must be preferentially allocated to this type of services to prevent packet loss.
OM MML services	The priority of this type of services is high.	OM services have small traffic volume and their transmission bandwidth must be preferentially guaranteed.
OM FTP services	The priority of this type of services is lower than those of other types of services.	For FTP services, the maximum transmission rate is 2400 kbit/s on a single connection. OM upload/download traffic fluctuates greatly. It is recommended that the QoS priority of such OM services, which have low real-time requirements, be lower than those of other services to prevent the impact of traffic burst on other services. However, a minimum bandwidth must be guaranteed for such OM services.
IP clock services: - Control packets - Clock packets	The priority of this type of services is high.	Clock synchronization signals have small traffic volume and their transmission bandwidth must be preferentially guaranteed.
Passing-by data flows	N/A	A base station schedules passing-by data flows based on their carried DSCPs.

Capacity Requirements

In terms of capacity, this feature is designed to:

- Admit as many services as possible without sacrificing service quality.
- Maximize bandwidth usage when data transmission is required to provide the highest possible throughput and to prevent congestion caused by non-real-time services, which are prone to data burst.

Differentiated Service Requirements

Differentiated service (DiffServ) is an important means to ensure the QoS of IP transmission. The QoS system provides different quality guarantees for different service types.

A base station can select different transmission bearer types and transmission priorities for these service types.

Table 4-2 describes differentiated service requirements for different types of services.

Table 4-2 Differentiated service requirements for different types of services

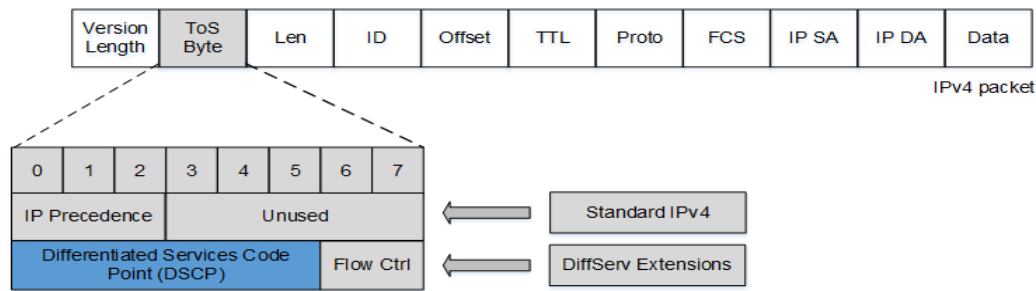
Control Type	Service Type	DiffServ Requirement	Description
Flow-controllable services	Non-real-time services	When the uplink transmission bandwidth of the base station is limited, the available bandwidth is preferentially allocated to non-flow-controllable services, and then the remaining available bandwidth is allocated to each service based on the scheduling priority weighting factor.	The traffic volume fluctuates significantly.
	OM FTP services	These services have lower priorities than other services. The minimum bandwidths or basic resources must be guaranteed for these services.	None
	eXn services	The priority of this type of services is lower than that of S1/X2/NG/Xn/SA X2 services.	None
Non-flow-controllable services	Real-time services	When a base station does not have enough bandwidth available in the uplink, the available bandwidth must be preferentially allocated to real-time services.	The total traffic volume fluctuates insignificantly when multiple services are admitted.
	Control-plane services	Related data must be preferentially transmitted during network congestion to ensure low packet loss rates and short delay.	None
	OM MML services	Related data must be preferentially transmitted during network congestion to ensure low packet loss rates and short delay.	None

Control Type	Service Type	DiffServ Requirement	Description
	IP clock services	Related data must be preferentially transmitted during network congestion to ensure low packet loss rates and short delay.	None

4.1.2.1.2 Mapping Between Service Types and DSCPs

IP transmission is used over the S1/X2/NG/Xn/eXn/SA X2 interface. DiffServ is used to ensure IP transmission quality. DiffServ enables an NE to inform each router on a transport path of the QoS requirements, which are indicated in the DSCP field in an IP packet header. **Figure 4-3** shows the structure of the DSCP field in an IPv4 packet. The DSCP value ranges from 0 to 63. A larger DSCP value indicates a higher scheduling priority for the packet.

Figure 4-3 Structure of the DSCP field in an IPv4 packet



With DiffServ, services are classified, flow control is performed, and the DSCP field in each IP packet is set, based on the QoS requirements of the services before they are processed on the transmission network. In a network, the QoS mechanism differentiates each type of service and its QoS requirements according to the DSCP field and provides services such as resource allocation, queue scheduling, and packet discarding policies. These services are collectively called per-hop behavior (PHB). Most nodes on the network, such as routers on the transport network, comply with PHBs based on the DSCP field in IP packets.

To meet DiffServ QoS requirements and take full advantage of the benefits provided by DiffServ, appropriate DSCP values must be set on the base station for uplink IP packets transmitted over the S1/X2/NG/Xn/eXn/SA X2 interface. These DSCP values are assigned based on the QoS requirements for each service type, and the DSCP values for downlink packets are set by the EPC or NGC.

The base station prioritizes different types of service packets transmitted by a base station interface board based on packet service types and their DSCPs. **Table 4-3** lists the default mapping between service types and DSCPs. If the QCI/5QI type is not listed in the table, configure the mapping between service types and the DSCPs based on the network plan.

Table 4-3 Default mapping between service types and DSCPs

Service Type	QCI/5QI	DSCP	Resource Type
User plane	1	46	GBR
	2	34	
	3	34	
	4	34	
	65	46	
	66	46	
	67	46	
	71	18	
	72	18	
	73	18	
	74	18	
	75	46	
	76	18	
	5	46	Non-GBR
	6	18	
	7	18	
	8	18	
	9	0	
	69	46	
	70	18	
	79	18	
	80	18	
	82	46	Delay-critical GBR
	83	46	
	84	46	
	85	46	
	86	46	
Control plane	-	48	-
OM (MML)	-	46	-

Service Type	QCI/5QI	DSCP	Resource Type
OM (FTP)	-	18	-
IP clock	-	46	-

 NOTE

The eXn user plane corresponds to QCI/5QI 9.

Signaling, OM data, and IP clock data are mapped to DSCP values using the **DIFPRI** MO. In co-MPT scenarios, the RAT-specific mapping between signaling and DSCP values can be separately configured using the **SCTPTEMPLATE** MO.

Parameters related to user data types can be configured using the **UDT** and **UDTPARAGRP** MOs. [Table 4-4](#) lists the required parameters. Note that the **UDT.UDTPARAGRPID** and **UDTPARAGRP.UDTPARAGRPID** parameters must be set to the same value.

Table 4-4 Parameters required for the transmission parameter group of a user data type

MO	Parameter ID	Configuration Item
UDT	UDT.UDTNO	Number of a user data type. By default, there is a one-to-one mapping between user data type numbers and QCIs. Operator-specific mappings between QCIs and UDTs can be modified using the gNBOperatorQciParam.Qci , gNBOperatorQciParam.OperatorId , and gNBOperatorQciParam.UDTNO parameters in gNBOperatorQciParam MOs. User data types are classified into standard user data types (automatically configured by the system) and extended user data types (manually configured by running the ADD UDT command).
	UDT.UDTPARAGRPID	ID of the transmission parameter group for a user data type. Values 40 to 48 are reserved for standard user data types and are not recommended for extended user data types.
UDTPARAGRP	UDTPARAGRP.UDTPARAGRPI D	ID of the transmission parameter group for a user data type. Values 40 to 48 are reserved for standard user data types and are not recommended for extended user data types.

MO	Parameter ID	Configuration Item
	UDTPARAGRP. PRIRULE	Priority rule
	UDTPARAGRP. PRI	Priority

DSCP values must be planned end to end to comply with rules for mapping DSCPs to PHBs on transmission equipment. A larger DSCP value indicates a higher priority when well planned.

 NOTE

- The mapping between QCIs and UDTs that takes effect for a roaming UE is determined by the **gNBOperatorQciParam** MOs configured for the operator that is actually serving the UE.
- When the **gNBOperatorQciParam.UDTNO** parameter is set to **65535**, there is a one-to-one mapping between QCIs and UDTNOs.

4.1.2.1.3 Mapping Between DSCPs and VLAN Priorities

The VLAN priority is specified by the packet priority indicator in a VLAN tag. The transport network provides differentiated services based on the VLAN priority.

VLAN priorities of packets with different service types are determined by DSCP values and different MOs are used to configure different models.

- Old model: the **DSCPMAP** MO is used for configuration.
- New model: the **DSCP2PCPMAP**, **DSCP2PCPREF**, and **INTERFACE** MOs are used for configuration.

Table 4-5 lists the default mapping between DSCPs and VLAN priorities.

Table 4-5 Default mapping between DSCPs and VLAN priorities

DSCP	VLAN Priority
0 to 7	0
8 to 15	1
16 to 23	2
24 to 31	3
32 to 39	4
40 to 47	5
48 to 55	6
56 to 63	7

The transmission board of the base station has a certain processing capability. When the external downlink service traffic exceeds the processing capability of the board, the packet buffer resources of the board may be exhausted. As a result, the board cannot receive and process packets and services are affected. In this case, the board is disconnected from what is managing it, and signaling transmission is interrupted.

To ensure that high-priority services can be processed properly when the board is overloaded (the downlink traffic exceeds 4 Gbit/s), the base station provides the priority-based downlink rate limiting function. This function can be enabled by turning on the switch specified by **TRANSFUNCTIONSW.DLLIMITSW**. When a board is overloaded, packets with low priorities are discarded to prevent packet buffer resources from being exhausted and to ensure that services with high priorities are processed properly.

Packet priority mapping rules are as follows:

- IP packets are mapped based on DSCP values. Packets with DSCP values greater than or equal to 32 are mapped to high priorities, and packets with DSCP values less than 32 are mapped to low priorities.
- Non-IP packets that carry VLAN tags are mapped based on VLAN priorities. Packets with VLAN priorities greater than or equal to 4 are mapped to high priorities, and packets with VLAN priorities less than 4 are mapped to low priorities.
- By default, non-IP packets that do not carry VLAN tags are mapped to low priorities.

 NOTE

Only the UMPTe supports priority-based downlink rate limiting.

4.2 Network Analysis

4.2.1 When to Use

The same transmission resource group can be specified for the S1/X2/NG/Xn/eXn/SA X2 interface.

A gNodeB cannot implement flow control on passing-by data in co-transmission scenarios. If local data and passing-by data share a transmission resource group and uplink bandwidth is insufficient, the passing-by data preempts bandwidth from the local data. It is recommended that different transmission resource groups be specified for local data and passing-by data in co-transmission scenarios. The transmission resource group scheduling algorithm is then used to ensure fair bandwidth allocation for local data and passing-by data.

In cascading scenarios, a separate transmission resource group must be configured for lower-level base stations so that they do not share the transmission resource group with the local base station.

Transmission resource mapping is supported by default and must be enabled.

4.2.2 Required Information

Transmission resource mapping requires an operator's transmission QoS plan, including QCIs/5QIs, DSCPs, and VLAN priorities.

Transmission resource mapping on the gNodeB side includes:

- Mapping of control-plane signaling, OM data, and IP clock packets to DSCPs
- Mapping of user-plane data to DSCPs
- Mapping of DSCPs to VLAN priorities

4.2.3 Benefits

Transmission resource configuration and mapping provide UEs with differentiated services based on service QoS requirements to ensure fairness.

4.2.4 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Transmission DiffServ QoS provides differentiated quality guarantee for different service types, with QoS requirements of high-priority services preferentially met.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Only the UMPTe supports priority-based downlink rate limiting. Other functions are supported by all NR-capable main control boards. To learn which main control boards are NR-capable, see technical specifications of the related BBUs in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

Users can modify the attributes of a default port transmission resource group only after the group has been added.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Transmission Resource Configuration

MOs related to transmission resource configuration include:

- **RSCGRP** (old model)/**IPRSCGRP** (new model)
This MO is the basis for user-plane data transmission and transmission resource management.
- **GTRANSPARA**
This MO is used to configure the rate configuration mode, which can be either single-rate or dual-rate.
- **UDT**, **UDTPARAGRP**, **DIFPRI**, and **SCTPTEMPLATE**
These MOs are used to configure the mapping between service types and transmission resources. Differentiated QoS is provided for different service types.

- **EP2RSCGRP** (old model)/**EP2IPRSCGRP** (new model)
In endpoint configuration mode, this MO is used to add an endpoint group containing the local and peer user-plane IP addresses to a user-defined transmission resource group. In this way, user-plane data is mapped to the transmission resource group.
- **TRANSFUNCTIONSW**
This MO defines the transmission function switch for the base station.
- **DSCPMAP** (old model)/**DSCP2PCPMAP** + **DSCP2PCPREF** + **INTERFACE** (new model)
This MO specifies the mapping between DSCPs and VLAN priorities of packets with different service types. The transport network provides differentiated services based on the VLAN priorities.

The following describes the parameters that must be set in these MOs.

- Global transport parameters
Set parameters in the **GTRANSPARA** MO to turn on the resource group scheduling weight switch and specify the rate configuration mode. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
Resource Group Scheduling Weight Switch	GTRANSPARA.LPSCHSW	Set this parameter based on the network plan.
Rate Config Type	GTRANSPARA.RATECFGTYPE	The default value is SINGLE_RATE . Set this parameter to DUAL_RATE in multi-operator sharing scenarios where more precise transmission resource management is required.

- Transmission resource group for user-plane data
Set parameters in an **RSCGRP** (old model)/**IPRSCGRP** (new model) MO to configure the transmission resource group for user-plane data for bandwidth management. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
Transmission Resource Group ID (old model)/IP Transmission Resource Group ID (new model)	RSCGRP.RSCGRPID (old model)/ IPRSCGRP.IPRSCGRPID (new model)	• If there is no special requirement for the default transmission resource group, the MO is not required. • If you need to measure the performance counters of a transmission resource group, this MO is required.

Parameter Name	Parameter ID	Setting Notes
Subboard Type	RSCGRP.SBT (old model)/ IPRSCGRP.SBT (new model)	Set this parameter to BASE_BOARD .
Bearing Port Type	RSCGRP.PT (old model)/ IPRSCGRP.PT (new model)	Set this parameter to ETH or ETHTRK .
Tx Bandwidth	RSCGRP.TXBW (old model)/ IPRSCGRP.TXBW (new model)	Set this parameter based on the network plan. This parameter specifies the maximum transmit bandwidth of a transmission resource group in single-rate mode for traffic shaping.
TX Committed Burst Size	RSCGRP.TXCBS (old model)/ IPRSCGRP.TXCBS (new model)	For a transmission resource group, the value of RSCGRP.TXCBS (old model)/ IPRSCGRP.TXCBS (new model) must be greater than or equal to that of RSCGRP.TXBW (old model)/ IPRSCGRP.TXBW (new model) for traffic shaping.
TX Excessive Burst Size	RSCGRP.TXEBS (old model)/ IPRSCGRP.TXEBS (new model)	Set this parameter to a value twice the value of RSCGRP.TXBW (old model)/ IPRSCGRP.TXBW (new model).
Scheduling Weight	RSCGRP.WEIGHT (old model)/ IPRSCGRP.WEIGHT (new model)	This parameter is used to calculate the bandwidth proportion of a resource group for user admission control. If physical resources are insufficient, set this parameter based on the network plan. The default value is recommended.
TX Committed Information Rate	RSCGRP.TXCIR (old model)/ IPRSCGRP.TXCIR (new model)	This parameter specifies the transmit committed information rate for a transmission resource group, which is allocated to the operator. Set this parameter based on the network plan. This parameter is used for uplink transmission resource group scheduling in dual-rate mode.
TX Peak Information Rate	RSCGRP.TXPIR (old model)/ IPRSCGRP.TXPIR (new model)	This parameter is used for uplink transmission resource group scheduling and traffic shaping in dual-rate mode.

Parameter Name	Parameter ID	Setting Notes
TX Peak Burst Size	RSCGRP.TXPBS (old model)/ IPRSCGRP.TXPBS (new model)	- For a transmission resource group, the value of RSCGRP.TXPBS (old model)/IPRSCGRP.TXPBS (new model) must be greater than or equal to that of RSCGRP.TXCBS (old model)/IPRSCGRP.TXCBS (new model). - For a transmission resource group, the value of RSCGRP.TXPBS (old model)/IPRSCGRP.TXPBS (new model) must be greater than or equal to that of RSCGRP.TXPIR (old model)/IPRSCGRP.TXPIR (new model).

- Differentiated service priorities

QoS priorities must be configured for different types of services. Priorities of signaling, OM data, and IP clock packets are configured using a **DIFPRI** MO. Parameters in this MO have default values, which can be modified, but this MO cannot be added or removed. In co-MPT scenarios, the RAT-specific signaling priority is configured in the **SCTPTEMPLATE** MO. QoS priorities of user-plane data are configured using the **UDT** and **UDTPARAGRP** MOs.

The following table describes the key parameters in the **DIFPRI** MO.

Parameter Name	Parameter ID	Setting Notes
Priority Rule	DIFPRI.PRIRULE	Set this parameter based on the network plan.
Signaling Priority	DIFPRI.SIGPRI	The default value is recommended.
OM High Priority	DIFPRI.OMHIGHPRI	The default value is recommended. If the value of this parameter needs to be changed, change it according to the network plan.
OM Low Priority	DIFPRI.OMLOWPRI	The default value is recommended. If the value of this parameter needs to be changed, change it according to the network plan.

Parameter Name	Parameter ID	Setting Notes
IP Clock Priority	DIFPRI.<i>IPCLKPRI</i>	The default value is recommended. If the value of this parameter needs to be changed, change it according to the network plan.

The following table describes the key parameters in an **SCTPTEMPLATE** MO.

Parameter Name	Parameter ID	Setting Notes
DSCP Switch	SCTPTEMPLATE.<i>DSCPSW</i>	Set this parameter based on the network plan.
DSCP	SCTPTEMPLATE.<i>DSCP</i>	The default value is recommended. If the value of this parameter needs to be changed, change it according to the network plan.

The following table describes the key parameters in a **gNBOperatorQciParam** MO.

Parameter Name	Parameter ID	Setting Notes
Operator ID	gNBOperatorQciParam.<i>OperatorId</i>	Set this parameter based on the network plan.
QoS Class Identifier	gNBOperatorQciParam.<i>Qci</i>	Set this parameter based on the network plan.
User Data Type Number	gNBOperatorQciParam.<i>UDTN</i>	The default value is used. If the value of this parameter needs to be changed, change it according to the network plan.

The following table describes the key parameters in a **UDT** MO.

Parameter Name	Parameter ID	Setting Notes
User Data Type Number	UDT.UDTNO	Set this parameter based on the network plan.
User Data Type Transfer Parameter Group ID	UDT.UDTPARAGRPID	Set this parameter based on the network plan.

The following table describes the key parameters in a **UDTPARAGRP** MO.

Parameter Name	Parameter ID	Setting Notes
User Data Type Transfer Parameter Group ID	UDTPARAGRP.UDTPARAGRPI D	Set this parameter to the value of UDT.UDTPARAGRPID .
Priority	UDTPARAGRP.PRI	Set this parameter based on the network plan.

The following table describes the key parameters in a **DSCPMAP** MO (old model).

Parameter Name	Parameter ID	Setting Notes
Differentiated Services Code Point	DSCPMAP.DSCP	-
VLAN Priority	DSCPMAP.VLANP RIO	-

The following table describes the key parameters in the **DSCP2PCPMAP**, **DSCP2PCPREF**, and **INTERFACE** MOs (new model).

Parameter Name	Parameter ID	Setting Notes
DSCP-to-PCP Mapping ID	DSCP2PCPMAP.D SCP2PCPMAPID	-
Differentiated Services Code Point	DSCP2PCPREF.DS CP	-
PCP	DSCP2PCPREF.PC P	-

Parameter Name	Parameter ID	Setting Notes
Interface ID	INTERFACE.ITFID	-
Interface Type	INTERFACE.ITFTYPE	-
Port Type	INTERFACE.PT	-
DSCP-to-PCP Mapping ID	INTERFACE.DSCP2PCPMAPID	-

- Mapping between endpoints and transmission resource groups
Configure the mapping between endpoints and transmission resource groups. If it is not configured, the default transmission resource group is used. If it is configured, the specified transmission resource group is used. Set parameters in the **EP2RSCGRP** (old model)/**EP2IPRSCGRP** (new model) MO. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
End Point ID	EP2RSCGRP.ENDPOINTID (old model)/ EP2IPRSCGRP.ENDPOINTID (new model)	Set this parameter based on the network plan.
Transmission Resource Group ID (old model)/IP Transmission Resource Group ID (new model)	EP2RSCGRP.RSCGRPID (old model)/ EP2IPRSCGRP.IPRSCGRPID (new model)	Set this parameter based on the network plan.

- Transmission function switch
Set parameters in the **TRANSFUNCTIONSW** MO to enable the downlink rate limiting function. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
Downlink Rate Limiting Switch	TRANSFUNCTIONSW.DLLI/MITSW	The default value is DISABLE . When a board is overloaded (the downlink traffic exceeds 4 Gbit/s), set this parameter to ENABLE to enable the priority-based downlink rate limiting function, ensuring that services with higher priorities can be processed properly.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

In endpoint configuration mode, run the **ADD EPGROUP**, **ADD USERPLANEHOST**, **ADD USERPLANEPEER**, **ADD UPHOST2EPGRP**, and **ADD UPPEER2EPGRP** commands to configure user-plane paths.

- For the configuration method in NSA networking, see *X2 and S1 Self-Management in NSA Networking*, and *gNodeB Initial Configuration Guide (MML-based)* in *3900 & 5900 Series Base Station Product Documentation*.
- For the configuration method in SA networking, see *NG and Xn Self-Management*, *X2 Self-Management in SA Networking*, and *gNodeB Initial Configuration Guide (MML-based)* in *3900 & 5900 Series Base Station Product Documentation*.

When the old transmission configuration model is used:

```
//Configuring the base station rate mode and the transmission resource group scheduling weight switch
SET GTRANS PARA: LPSCHSW=ENABLE, RATECFGTYPE=SINGLE_RATE;
//Adding transmission resource groups for easy user-plane resource management
ADD RSCGRP: SN=7, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=0, RU=KBPS, TXBW=100000,
TXCBS=120000, TXEBS=120000, TXCIR=80000, TXPIR=100000, TXPBS=120000;
ADD RSCGRP: SN=7, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=1, RU=KBPS, TXBW=100000,
TXCBS=120000, TXEBS=120000, TXCIR=80000, TXPIR=100000, TXPBS=120000;
//Adding the mapping between endpoint groups and transmission resource groups
ADD EP2RSCGRP: MT=ENDPOINT_GROUP, ENDPOINTID=0, SN=7, SBT=BASE_BOARD, PT=ETH, RSCGRPID=0;
ADD EP2RSCGRP: MT=ENDPOINT_GROUP, ENDPOINTID=1, SN=7, SBT=BASE_BOARD, PT=ETH, RSCGRPID=1;
```

```
//Setting the priorities for control-plane data, OM data, and IP clock data
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=48, OMLOWPRI=14, IPCLKPRI=48;
//((Optional) Setting the RAT-specific DSCP for signaling packets in endpoint mode in co-MPT scenarios
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;
//Setting the priority for user-plane data
MOD UDT: UDTNO=1, UDTParagRPID=40;
MOD UDTParagRP: UDTParagRPID=40, PRIRULE=DSCP, PRI=46;
//Adding a user data type
ADD UDT: UDTNO=11, UDTParagRPID=40;
//((Optional) Turning on the downlink rate limiting switch when the board is overloaded (the downlink
traffic exceeds 4 Gbit/s)
SET TRANSFUNCTIONSW: DLLIMITSW=ENABLE;
//Adding a mapping between QCI and user data type numbers
ADD GNBOPERATORQCIPARAM: OperatorId=0, Qci=12, UDTNO=11;
//Setting the mapping between DSCPs and VLAN priorities
SET DSCPMap: DSCP=0, VLANPRI=7;
```

When the new transmission configuration model is used:

```
//Configuring the base station rate mode and the transmission resource group scheduling weight switch
SET GTRANSPARA: LPSCHSW=ENABLE, RATECFGTYPE=SINGLE_RATE;
//Adding transmission resource groups for easy user-plane resource management
ADD IPRSCGRP: IPRSCGRPID=0, PT=ETH, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=100000,
TXCBS=120000, TXEBS=120000, TXCIR=80000, TXPIR=100000, TXPBS=120000;
ADD IPRSCGRP: IPRSCGRPID=1, PT=ETH, PORTID=0, RSCGRPNO=1, RU=KBPS, TXBW=100000,
TXCBS=120000, TXEBS=120000, TXCIR=80000, TXPIR=100000, TXPBS=120000;
//Adding the mapping between endpoint groups and transmission resource groups
ADD EP2IPRSCGRP: MAPID=0, MT=ENDPOINT_GROUP, ENDPOINTID=0, IPRSCGRPID=0;
ADD EP2IPRSCGRP: MAPID=1, MT=ENDPOINT_GROUP, ENDPOINTID=1, IPRSCGRPID=1;
//Setting the priorities for control-plane data, OM data, and IP clock data
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=48, OMLOWPRI=14, IPCLKPRI=48;
//((Optional) Setting the RAT-specific DSCP for signaling packets in endpoint mode in co-MPT scenarios
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;
//Setting the priority for user-plane data
MOD UDT: UDTNO=1, UDTParagRPID=40;
MOD UDTParagRP: UDTParagRPID=40, PRIRULE=DSCP, PRI=46;
//Adding a user data type
ADD UDT: UDTNO=11, UDTParagRPID=40;
//((Optional) Turning on the downlink rate limiting switch when the board is overloaded (the downlink
traffic exceeds 4 Gbit/s)
SET TRANSFUNCTIONSW: DLLIMITSW=ENABLE;
//Adding a mapping between QCI and user data type numbers
ADD GNBOPERATORQCIPARAM: OperatorId=0, Qci=12, UDTNO=11;
//Setting the mapping between DSCPs and VLAN priorities
ADD DSCP2PCPMap: DSCP2PCPMapID=1;
ADD DSCP2PCPREF: DSCP2PCPMapID=1, DSCP=0, PCP=7;
MOD INTERFACE: ITFID=1, ITFTYPE=VLAN, PT=ETH, DSCP2PCPMapID=1;
```

Deactivation Command Examples

In endpoint configuration mode, remove the mapping between the endpoint group containing user-plane data and the dedicated transmission resource group.

When the old transmission configuration model is used:

```
RMV EP2RSCGRP: MT=ENDPOINT_GROUP, ENDPOINTID=0, SN=7, SBT=BASE_BOARD, PT=ETH, RSCGRPID=0;
RMV EP2RSCGRP: MT=ENDPOINT_GROUP, ENDPOINTID=1, SN=7, SBT=BASE_BOARD, PT=ETH, RSCGRPID=1;
```

When the new transmission configuration model is used:

```
RMV EP2IPRSCGRP: MAPID=0;
RMV EP2IPRSCGRP: MAPID=1;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Note that:

- An IP layer protocol tracing task must be created and started on the MAE-Access.
- The methods used to access the network and set up a dedicated bearer depend on the types of UEs. For details, see the user guide provided by the UE manufacturer.
- The methods used to perform packet injection in the uplink and downlink depend on the packet injection tools and data types. UDP packet injection is used as an example in this section.

Prerequisites

The prerequisites for transmission resource configuration and mapping are as follows:

- The cell status is normal.
- The base station uses the single-rate mode.
- A default bearer with a QCI/5QI of 9 has been set up for the test UE. The QCI/5QI for the default bearer is determined by the core network upon UE registration.

Unless otherwise specified, activation verification is performed in single-rate mode. To set the rate configuration mode, run the **SET GTRANSPARA** command.

Procedure

- Step 1** Enable a UE with the default bearer to access the cell. Perform UDP packet injection in the uplink/downlink.
- Step 2** Check whether the DSCP value of packets sent from the base station is the same as the DSCP values configured in the **DIFPRI/SCTPTEMPLATE** and **UDTPARAGRP** MOs.
1. Create an IP trace task. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**, and then in the navigation tree of the **Signaling Trace Management** tab page, choose **Trace Type > Base Station Device and Transport > Transport Trace > IP Layer Protocol Trace**. In the tracing result, the **Type Of Service** value for SCTP packets is 184. According to the mapping between service types and DSCPs, the DSCP value for SCTP packets is 46 (184/4).
 2. Run the **LST DIFPRI/LST SCTPTEMPLATE** and **LST UDTPARAGRP** commands to view the DSCP values for user- and control-plane data. The command output shows that the queried DSCP values are the same as those obtained from the IP trace task, which indicates that the DSCP settings have taken effect.

----End

4.4.3 Network Monitoring

None

5 Transmission Congestion Control

5.1 Principles

Transmission congestion control algorithms improve the quality of the transmission network when there is congestion in a base station due to frequent transmission quality fluctuations.

Transmission congestion control algorithms consist of the following:

- Traffic shaping
The transmit rate for traffic shaping of a transmission resource group cannot exceed any bandwidth bottlenecks on the network, preventing congestion on the transmission network.
Rate limiting on physical ports ensures that the transmit rate of a physical port does not exceed the limited bandwidth of the physical port.
- Queue scheduling
Weight-based service scheduling is used so that each service has a chance to be scheduled.
- Backpressure algorithm
The transmit rate of uplink non-real-time services is limited to prevent congestion of transmission resource groups.

5.1.1 Base Station Transmission Congestion Control

A base station performs congestion control on uplink data. Congestion control involves traffic shaping, queue scheduling, and a backpressure algorithm.

If the bandwidth over the S1/X2/NG/Xn/eXn interface is insufficient, there will be congestion. The amount of data to be transmitted will exceed the capacity of the available bandwidth. The SA X2 interface is used only to transmit control-plane packets and typically does not experience congestion.

The S1/X2/NG/Xn/eXn interface becomes congested when:

- The traffic on the S1/NG interface is heavy, and excessive services are admitted.

- The X2/Xn interface is mainly used to transmit handover data. Although the average traffic is light, it has frequent burst traffic and sometimes heavy transient traffic.
- The eXn interface is used to transmit inter-base-station coordination data and has heavy traffic.

Congestion control can provide users with differentiated services and guarantee fairness.

- Differentiated services: When no sufficient bandwidth is available, the backpressure algorithm is used to preferentially ensure the required bandwidth for non-flow-controllable services. Non-real-time services are differentiated based on priorities. High-priority users are prioritized over low-priority users. The available bandwidth is allocated among users based on their weighting factors when the minimum committed information rate (specified by **gNBFlowCtrlParamGrp.UlFlowCtrlMinCir**) is ensured.
- Fairness: When congestion occurs, packet loss occurs on the interface boards, which affects fairness among non-real-time services. For example, a service that has established multiple TCP connections will preempt bandwidth from services that have the same QCI/5QI but only a single TCP connection. In this case, the backpressure algorithm can be used to ensure that each admitted user has certain bandwidth based on the weighting factor to prevent resource shortages. The weighting factor is specified by the **gNBFlowCtrlParamGrp.UlFlowCtrlPriWeightFactor** parameter.

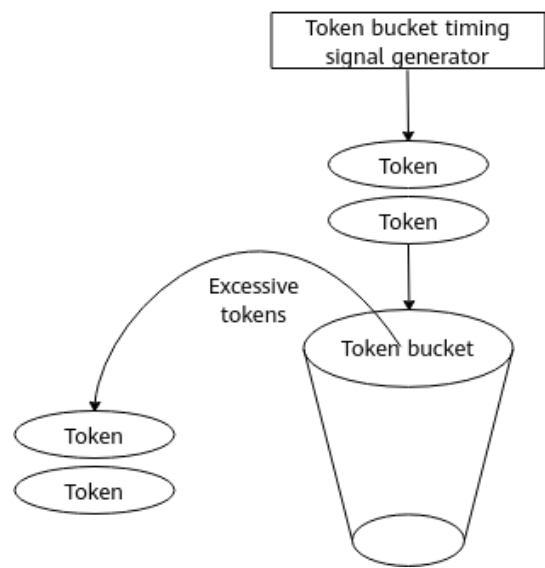
5.1.1.1 Traffic Shaping

When the network is congested, the traffic shaping function can be used to limit traffic and reduce packet loss. It is implemented at the data link layer.

Traffic shaping aims to limit outgoing traffic and data bursts on a network connection, maintaining steady packet transmission rates. This function uses generic traffic shaping (GTS) technology to transform irregular traffic patterns or those that do not comply with predefined characteristics, facilitating bandwidth matching between upstream and downstream networks.

GTS operates on the token bucket principle, as shown in [Figure 5-1](#).

Figure 5-1 Token bucket principle



The token bucket size is the maximum number of tokens that can be buffered. Tokens are periodically generated by the token bucket timing signal generator and added to the token bucket. When there are no tokens in the token bucket, packet transmission is not allowed. When the token bucket is full, new tokens will be discarded.

Base stations implement two levels of traffic shaping based on the token bucket principle: traffic shaping of transmission resource groups and rate limiting on physical ports. These two levels of traffic shaping make flow control possible.

Traffic Shaping of Transmission Resource Groups

Traffic shaping of transmission resource groups is a rate limiting mechanism. It is controlled by the **RSCGRP.TXSSW** (old model)/**IPRSCGRPALG.TXSSW** (new model) parameter. This mechanism prevents the transmission rate of a transmission resource group from exceeding the bandwidth configured for this group.

- **Table 5-1** describes the parameters related to traffic shaping of transmission resource groups in single-rate mode.

Table 5-1 Parameters related to traffic shaping of transmission resource groups in single-rate mode

Parameter Name	Parameter ID	Description
Tx Bandwidth	RSCGRP.TXBW (old model)/ IPRSCGRP.TXBW (new model)	This parameter specifies the token injection rate.

Parameter Name	Parameter ID	Description
TX Committed Burst Size	RSCGRP.TXCBS (old model)/ IPRSCGRP.TXCBS (new model)	RSCGRP.TXCBS (old model)/ IPRSCGRP.TXCBS (new model) and RSCGRP.TXEBS (old model)/ IPRSCGRP.TXEBS (new model) determine the token bucket size.
TX Excessive Burst Size	RSCGRP.TXEBS (old model)/ IPRSCGRP.TXEBS (new model)	

- **Table 5-2** describes the parameters related to traffic shaping of transmission resource groups in dual-rate mode.

Table 5-2 Parameters related to traffic shaping of transmission resource groups in dual-rate mode

Parameter Name	Parameter ID	Description
TX Committed Information Rate	RSCGRP.TXCIR (old model)/ IPRSCGRP.TXCIR (new model)	This parameter specifies the token injection rate.
TX Committed Burst Size	RSCGRP.TXCBS (old model)/ IPRSCGRP.TXCBS (new model)	RSCGRP.TXCBS (old model)/ IPRSCGRP.TXCBS (new model) and RSCGRP.TXEBS (old model)/ IPRSCGRP.TXEBS (new model) determine the token bucket size.
TX Excessive Burst Size	RSCGRP.TXEBS (old model)/ IPRSCGRP.TXEBS (new model)	
TX Peak Information Rate	RSCGRP.TXPIR (old model)/ IPRSCGRP.TXPIR (new model)	This parameter specifies the token injection rate.
TX Peak Burst Size	RSCGRP.TXPBS (old model)/ IPRSCGRP.TXPBS (new model)	This parameter specifies the token bucket size.

Rate Limiting on Physical Ports

Line-Rate (LR) can be used to limit the rate of all packets sent from an interface. If an **LR** (old model)/**PORTLR** (new model) MO is configured for a physical port, all the packets sent from this port must be processed using the LR token bucket for flow control regardless of the type of data flow. If there are tokens in the token bucket, burst transmission of packets is allowed to achieve both flow control

and transmission of burst traffic. [Table 5-3](#) describes the parameters related to rate limiting on physical ports.

Table 5-3 Parameters related to rate limiting on physical ports

Parameter Name	Parameter ID	Description
UL Committed Information Rate	LR.CIR (old model)/ PORTLR.CIR (new model)	This parameter specifies the token injection rate and the recommended value is 0 .
Committed Burst Size	LR.CBS (old model)/ PORTLR.CBS (new model)	LR.CBS (old model)/PORTLR.CBS (new model) and LR.EBS (old model)/PORTLR.EBS (new model) determine the token bucket size. - It is recommended that the value of LR.CBS (old model)/PORTLR.CBS (new model) be greater than or equal to the value of LR.CIR (old model)/PORTLR.CIR (new model).
Excessive Burst Size	LR.EBS (old model)/ PORTLR.EBS (new model)	

If the buffer size of the peer device can reach 1.5 to 2 times the value of **LR.CIR** (old model)/**PORTLR.CIR** (new model), it is recommended that **LR.CBS** (old model)/**PORTLR.CBS** (new model) be set to a value 1.5 to 2 times the value of **LR.CIR** (old model)/**PORTLR.CIR** (new model). If the buffer size of the peer device is less than 1.5 times the value of **LR.CIR** (old model)/**PORTLR.CIR** (new model), it is recommended that **LR.CBS** (old model)/**PORTLR.CBS** (new model) be set to the same value as the buffer size of the peer device.

5.1.1.2 Queue Scheduling

Queue scheduling ensures that non-flow-controllable services (including real-time services, control-plane services, OM MML services, and IP clock services) are scheduled preferentially.

Each transmission resource group for a gNodeB can be configured with up to seven queues. There are two types of queues:

- PQ queues are numbered from 0 to **RSCGRPALG.PQN** (old model)/**IPRSCGRPALG.PQN** (new model) minus 1. The number of PQ queues is indicated by **RSCGRPALG.PQN** (old model)/**IPRSCGRPALG.PQN** (new model).
- Non-PQ queues are numbered from **RSCGRPALG.PQN** (old model)/**IPRSCGRPALG.PQN** (new model) to 6

Queues in a transmission resource group are scheduled as follows:

1. PQ queues are preferentially scheduled. PQ queues with smaller IDs have higher scheduling priorities. PQ queues with low priorities are scheduled only when those with high priorities have no buffered data.
2. When all PQ queues have been scheduled, WRR-based scheduling is performed on non-PQ queues. All non-PQ queues have the same scheduling weight.

When the old transmission configuration model is used, service packets enter queues based on their DSCPs, which are mapped to service types. There is a mapping between service types and queues. The lowest priority of queue *x* is specified by the **PRI*x*** parameter, where *x* is a value from 0 to 6. For example, **PRI2QUE.PRI3** indicates the lowest priority of queue 3.

When the new transmission configuration model is used, the priority of a transmission resource group queue is configured using the **DSCP2QUEMAP** MO. The **DSCP** and **QUE** subparameters of the **DSCP2QUEMAP.DSCP2QUEREFS** parameter are used to set the mapping between DSCPs and queue priorities.

Table 5-4 describes the default mapping between service types and queues.

Table 5-4 Default mapping between service types and queues

Service Type	QCI/5QI	Queue ID	Resource Type
User plane	1	1	GBR
	2	2	
	3	2	
	4	2	
	65	1	
	66	1	
	67	1	
	71	4	
	72	4	
	73	4	
	74	4	
	75	1	
	76	4	
	5	1	
	6	4	
	7	4	
	8	4	Non-GBR
	9	6	

Service Type	QCI/5QI	Queue ID	Resource Type
	69	1	Delay-critical GBR
	70	4	
	79	4	
	80	4	
	82	1	
	83	1	
	84	1	
	85	1	
	86	1	
Control plane	-	0	-
OM (MML)	-	1	-
OM (FTP)	-	4	-
IP clock	-	1	-

5.1.1.3 Backpressure Algorithm

The backpressure algorithm limits the transmit rates of uplink non-real-time services to prevent congestion of transmission resource groups. The backpressure algorithm is controlled by the **RSCGRPALG.TCSW** parameter. A base station does not implement backpressure on real-time services or passing-by data flows.

The backpressure algorithm consists of two parts:

- The interface board detects that a transmission resource group is congested and sends a backpressure signal to the service board.
- The service board buffers data of each non-real-time service separately and adjusts the maximum transmit rate of each service.

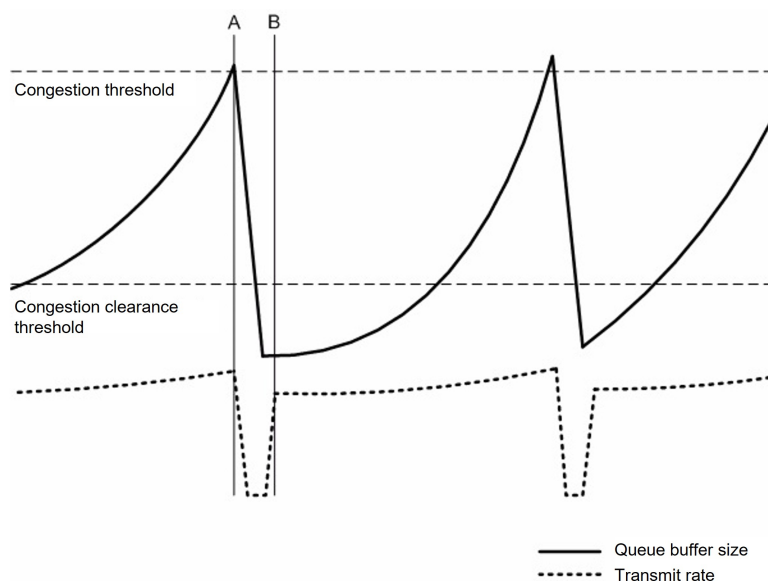
NOTE

The initial transmit rate is the UE-AMBR multiplied by 1.25. The UE-AMBR value is sent by the MME to the base station. The backpressure algorithm limits the transmit rate of services. Therefore, when the backpressure algorithm is enabled, the actual effect of the scheduling weight of transmission resource groups may be affected.

When LTE and NR do not share a transmission resource group, the backpressure algorithm does not take effect for non-real-time services in Option 3x if the LTE transmission resource group is congested but the NR transmission resource group is not congested.

When LTE and NR do not share a transmission resource group, the backpressure algorithm does not take effect for non-real-time services in Option 3 if the NR transmission resource group is congested but the LTE transmission resource group is not congested.

Figure 5-2 Backpressure process of non-real-time services



The flow control process using the backpressure algorithm shown in [Figure 5-2](#) is as follows:

1. The interface board periodically checks the buffer size of each queue in the transmission resource group.
2. When the buffer time of data in a queue exceeds the congestion threshold (**RSCGRPALG.CTTH**) or the number of data packets buffered in a queue exceeds 75% of the number of the data packets that can be buffered in the queue, the queue and the corresponding transmission resource group enter the congestion state, as shown by point A in the preceding figure. The interface board then sends congestion signals to the service board. The service board stops transmitting the data for all non-real-time services in the transmission resource group to decrease the transmit rate of each non-real-time service.
3. If the buffer size of a queue reaches the maximum value (**RSCGRPALG.DROPPKTNUM**), any new data packets that arrive are discarded.
4. When the buffer time of the queue drops below the congestion clearance threshold (**RSCGRPALG.CCTH**) and the data packets buffered in the queue are fewer than 50% of the data packets that can be buffered in the queue, the queue enters the congestion clearance state, as shown by point B in the preceding figure. If all the queues in the transmission resource group enter the congestion clearance state, the transmission resource group enters the congestion clearance state. The interface board then sends congestion clearance signals to the service board. The service board then transmits the data for non-real-time services in the transmission resource group at a rate that does not exceed the maximum transmit rate.
5. In the congestion clearance state, the backpressure algorithm periodically increases the transmit rate by the rate increase step, which is the difference between the transmit rates before and after the rate increase.

 NOTE

The rate increase step of each service has a positive correlation with the weighting factor specified by **gNBFlowCtrlParamGrp.UlFlowCtrlPriWeightFactor**.

To avoid impacts of eXn services on S1/X2/NG/Xn/SA X2 services, the system preferentially implements backpressure on eXn services and then on S1/X2/NG/Xn/SA X2 services, when transmission resource groups become congested.

When the backpressure algorithm switch **RSCGRPALG.TCSW** (old model)/**IPRSCGRPALG.TCSW** (new model) is set to **ENABLE**, the following configurations can be used to prevent gNodeB internal congestion in the uplink if the uplink rate of UEs needs to be limited over the air interface:

- In SA networking, select the **UL_UU_FLOW_CTRL_SW** option of the **gNodeBParam.UlFlowCtrlAlgoSwitch** parameter.
- In NSA networking, set parameters according to [Table 5-5](#).

Table 5-5 Configurations in NSA networking

Data Split Architecture for UE Flow Control Services	Uplink Air Interface Flow Control Parameter Configuration
Option 3x	Select both the UL_UU_FLOW_CTRL_SW and UL_UU_FLOW_CTRL_ENHANCED_SW options of the gNodeBParam.UlFlowCtrlAlgoSwitch parameter on the NR side. In dynamic uplink data split scenarios, the total uplink rate of UEs over the air interface on both the LTE and NR sides is limited.
Option 3	Select both the ULUuFlowCtrlSwitch and ULUuFlowCtrlEnhancedSwitch options of the ENodeBAlgoSwitch.TrmSwitch parameter on the LTE side. In dynamic uplink data split scenarios, the total uplink rate of UEs over the air interface on both the LTE and NR sides is limited.

5.2 Network Analysis

5.2.1 When to Use

Traffic shaping, queue scheduling, and backpressure algorithm are enabled by default.

When the transmission bandwidth planned by an operator is small, rate limiting on ports and traffic shaping of transmission resource groups can be configured on the gNodeB to avoid packet loss caused by congestion on the transmission link.

5.2.2 Benefits

Transmission congestion control can control and adjust the traffic volume when the transmission network is congested or the transmission network quality fluctuates frequently. This provides differentiated services and ensures fairness for UEs.

5.2.3 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

The MOs related to transmission congestion control are **RSCGRP** (old model)/**IPRSCGRP** (new model), **RSCGRPALG** (old model)/**IPRSCGRPALG** (new model), **LR** (old model)/**PORTLR** (new model), and **PRI2QUE** (old model)/**DSCP2QUEMAP** (new model).

Configuring Traffic Shaping of Transmission Resource Groups

Set parameters in the **RSCGRPALG** (old model)/**IPRSCGRPALG** (new model) MO to configure the traffic shaping switch for a transmission resource group. This MO is automatically generated after an **RSCGRP** (old model)/**IPRSCGRP** (new model) MO is configured. Before modifying parameter settings in this MO, ensure that the **RSCGRP** (old model)/**IPRSCGRP** (new model) MO has been configured. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
TX Traffic Shaping Switch	RSCGRPALG.TXSSW (old model)/ IPRSCGRPALG.TXSSW (new model)	Set this parameter based on the network plan. The value ON is recommended.

Configuring Rate Limiting on Physical Ports

Set parameters in the **LR** (old model)/**PORTLR** (new model) MO to configure rate limiting on a physical port and achieve traffic shaping on this physical port. This

MO can only be modified and cannot be added or removed. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
LR Switch	LR.LRSW (old model)	When the transmission network bandwidth is insufficient, it is necessary to limit the data rate on physical ports to prevent network congestion and packet loss.
UL Committed Information Rate	LR.CIR (old model)/ PORTLR.CIR (new model)	This parameter takes effect only when the LR switch is turned on. The LR.CIR (old model)/ PORTLR.CIR (new model) parameter is used to configure uplink traffic shaping for physical ports.
Committed Burst Size	LR.CBS (old model)/ PORTLR.CBS (new model)	This parameter takes effect only when the LR switch is turned on. It is recommended that the value of LR.CBS (old model)/ PORTLR.CBS (new model) be 1.5 to 2 times the value of LR.CIR (old model)/ PORTLR.CIR (new model).
Excess Burst Size (old model)/Excessive Burst Size (new model)	LR.EBS (old model)/ PORTLR.EBS (new model)	Set this parameter based on the network plan. This parameter takes effect only when the LR switch is turned on.

Configuring Queue Scheduling of Transmission Resource Groups

The **DIFPRI/SCTPTEMPLATE** and **PRI2QUE** (old model)/**DSCP2QUEMAP** (new model) MOs collectively specify the mapping between signaling priorities and scheduling queues, and the mapping between OM packet priorities and scheduling queues. The **UDTPARAGRP** and **PRI2QUE** (old model)/**DSCP2QUEMAP** (new model) MOs collectively specify the mapping between service packet priorities and scheduling queues. The **UDTPARAGRP** MO specifies the mapping between service types and DSCPs. The **PRI2QUE** (old model)/**DSCP2QUEMAP** (new model) MO specifies the mapping between DSCPs and queues.

Configure the mapping between transmission bearer priorities (only DSCPs are supported currently) and queues in a **PRI2QUE** MO when the old transmission configuration model is used. This MO can only be modified and cannot be added or removed. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
PriOfQue0	PRI2QUE.PRI0	This parameter specifies the DSCP priority of queue 0. Set this parameter based on the network plan. The default value is recommended.
PriOfQue1	PRI2QUE.PRI1	This parameter specifies the DSCP priority of queue 1. Set this parameter based on the network plan. The default value is recommended.
PriOfQue2	PRI2QUE.PRI2	This parameter specifies the DSCP priority of queue 2. Set this parameter based on the network plan. The default value is recommended.
PriOfQue3	PRI2QUE.PRI3	This parameter specifies the DSCP priority of queue 3. Set this parameter based on the network plan. The default value is recommended.
PriOfQue4	PRI2QUE.PRI4	This parameter specifies the DSCP priority of queue 4. Set this parameter based on the network plan. The default value is recommended.
PriOfQue5	PRI2QUE.PRI5	This parameter specifies the DSCP priority of queue 5. Set this parameter based on the network plan. The default value is recommended.
PriOfQue6	PRI2QUE.PRI6	This parameter specifies the DSCP priority of queue 6. Set this parameter based on the network plan. The default value is recommended.

The priority of queue 7 in the **PRI2QUE** MO cannot be configured and is always 0. Service packets with priorities lower than **PRI2QUE.PRI6** are all assigned to queue 7.

Configure the mapping between transmission bearer priorities (only DSCPs are supported currently) and internal queues in a **DSCP2QUEMAP** MO when the new transmission configuration model is used. This MO with **DSCP2QUEMAP.DSCP2QUEMAPID** set to **0** can only be modified and cannot be added or removed. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
Differentiated Services Code Point	DSCP2QUEREF.DSCP	This parameter specifies the priority corresponding to a DSCP value. Set this parameter based on the network plan. The default value is recommended.
Queue No.	DSCP2QUEREF.QUE	This parameter specifies the number of a queue. Set this parameter based on the network plan. The default value is recommended.

Configure the number of PQ queues in an **RSCGRPALG** (old model)/**IPRSCGRPALG** (new model) MO. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
PQ Number	RSCGRPALG.PQN (old model)/ IPRSCGRPALG.PQN (new model)	This parameter specifies the number of priority queues. Set this parameter based on the network plan. The default value is recommended.

Configuring the Backpressure Algorithm

Turn on the backpressure algorithm switch and set related parameters in an **RSCGRPALG** (old model)/**IPRSCGRPALG** (new model) MO. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Setting Notes
Traffic Control Switch	RSCGRPALG.TCSW (old model)/ IPRSCGRPALG.TCSW (new model)	Set this parameter based on the network plan. The default value is ENABLE .

Parameter Name	Parameter ID	Setting Notes
Congestion Time Threshold	RSCGRPALG.CTTH (old model)/ IPRSCGRPALG.CTTH (new model)	Set these parameters based on the network plan. Retain the default values unless there are special requirements. The value of RSCGRPALG.CCTTH (old model)/ IPRSCGRPALG.CCTTH (new model) must be less than the value of RSCGRPALG.CTTH (old model)/ IPRSCGRPALG.CTTH (new model).
Congestion Clear Time Threshold	RSCGRPALG.CCTTH (old model)/ IPRSCGRPALG.CCTTH (new model)	

(Optional) To limit the uplink rate of UEs, turn on the uplink flow control algorithm switch by using the **gNodeBParam** MO. The key parameters in this MO are listed in the following table.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Flow Control Algorithm Switch	gNodeBParam.UlFlowCtrlAlgoSwitch	- UL_UU_FLOW_CTRL_SW - UL_UU_FLOW_CTRL_ENHANCED_SW	Set this parameter based on the network plan. By default, the uplink flow control algorithms are disabled.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Configure traffic shaping, queue scheduling, and backpressure algorithm of a transmission resource group.

When the old transmission configuration model is used:

```
//Turning on the transmit traffic shaping switch and backpressure algorithm switch
SET RSCGRPALG: SN=7, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=0, TXSSW=ON,
```

```
TCSW=ENABLE;  
//(Optional) Enabling uplink flow control (which is disabled by default)  
//Enabling uplink flow control over the air interface in SA networking  
MOD GNODEBPARAM: UFlowCtrlAlgoSwitch=UL_UU_FLOW_CTRL_SW-1;  
//Enabling uplink flow control over the air interface and enhanced uplink flow control over the air  
interface in NSA networking  
MOD GNODEBPARAM:  
UFlowCtrlAlgoSwitch=UL_UU_FLOW_CTRL_SW-1&UL_UU_FLOW_CTRL_ENHANCED_SW-1;  
//Configuring priorities  
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=48, OMLOWPRI=14, IPCLKPRI=48;  
//(Optional) Setting the RAT-specific DSCP for signaling packets in endpoint mode in co-MPT scenarios  
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;  
//Modifying a transmission parameter group for a user data type  
MOD UDTPARAGRP: UDTPARAGRPID=48, PRIRULE=DSCP, PRI=0;  
//Configuring the mapping between priorities and internal queues  
SET PRI2QUE: PRI0=48, PRI1=40;  
//Configuring the number of PQ queues  
SET RSCGRPAlg: SN=7, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=0, PQN=3;
```

When the new transmission configuration model is used:

```
//Turning on the transmit traffic shaping switch and backpressure algorithm switch  
SET IPRSCGRPAlg: IPRSCGRPID=0, TXSSW=ON, TCSW=ENABLE;  
//(Optional) Enabling uplink flow control (which is disabled by default)  
//Enabling uplink flow control over the air interface in SA networking  
MOD GNODEBPARAM: UFlowCtrlAlgoSwitch=UL_UU_FLOW_CTRL_SW-1;  
//Enabling uplink flow control over the air interface and enhanced uplink flow control over the air  
interface in NSA networking  
MOD GNODEBPARAM:  
UFlowCtrlAlgoSwitch=UL_UU_FLOW_CTRL_SW-1&UL_UU_FLOW_CTRL_ENHANCED_SW-1;  
//Configuring priorities  
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=48, OMLOWPRI=14, IPCLKPRI=48;  
//(Optional) Setting the RAT-specific DSCP for signaling packets in endpoint mode in co-MPT scenarios  
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;  
//Modifying a transmission parameter group for a user data type  
MOD UDTPARAGRP: UDTPARAGRPID=48, PRIRULE=DSCP, PRI=0;  
//Configuring the mapping between priorities and internal queues  
ADD DSCP2QUEREF: DSCP2QUEMAPID=0, DSCP=48, QUE=0;  
ADD DSCP2QUEREF: DSCP2QUEMAPID=0, DSCP=40, QUE=1;  
//Configuring the number of PQ queues  
SET IPRSCGRPAlg: IPRSCGRPID=0, PQN=3;
```

- Configure rate limiting on a physical port.

When the old transmission configuration model is used:

```
SET LR: SN=7, SBT=BASE_BOARD, PT=ETH, PN=0, LRSW=ENABLE, CIR=100000, CBS=200000,  
EBS=150000;
```

When the new transmission configuration model is used:

```
ADD PORTLR: PORTLRID=0, PT=ETH, PORTID=0, CIR=100000, CBS=200000, EBS=150000;
```

Deactivation Command Examples

- Deactivate traffic shaping and backpressure algorithm of a transmission resource group.

When the old transmission configuration model is used:

```
SET RSCGRPAlg: SN=7, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=0, TXSSW=OFF,  
TCSW=DISABLE;
```

When the new transmission configuration model is used:

```
SET IPRSCGRPAlg: IPRSCGRPID=0, TXSSW=OFF, TCSW=DISABLE;
```

- Deactivate rate limiting on a physical port.

When the old transmission configuration model is used:

```
SET LR: SN=7, SBT=BASE_BOARD, PT=ETH, PN=0, LRSW=DISABLE;
```

When the new transmission configuration model is used:

```
RMV PORTLR: PORTLRID=0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Note that:

- An S1/NG tracing task must be created and started on the MAE-Access.
- The methods used to access the network and set up a dedicated bearer depend on the types of UEs. For details, see the user guide provided by the UE manufacturer.
- The methods used to perform packet injection in the uplink and downlink depend on the packet injection tools and data types. UDP packet injection is used as an example in this section.

Traffic Shaping of Transmission Resource Groups

Prerequisites

The committed bandwidth of a physical port is greater than the sum of the bandwidths of all transmission resource groups configured for the port.

Procedure

- Step 1** Run the **LST RSCGRP** (old model)/**LST IPRSCGRP** (new model) command to check the bandwidth settings of transmission resource groups.
- Step 2** Run the **LST RSCGRPALG** (old model)/**LST IPRSCGRPALG** (new model) command to check the settings of the traffic shaping switch and backpressure algorithm switch.
- Step 3** Run the **LST GNBFLOWCTRLPARAMGRP** command to check the uplink flow control priority weight factors for QCI/5QIs 6 and 8, and record the query results.
- Step 4** Run the **MOD GNBFLOWCTRLPARAMGRP** command to change the uplink flow control priority weight factors for QCI/5QIs 6 and 8 to **1000** and **500**, respectively.
- Step 5** Enable UE 1 and UE 2 to access the cell. Enable UE 1 to set up a non-GBR bearer with a QCI/5QI of 6, and enable UE 2 to set up a non-GBR bearer with a QCI/5QI of 8. Then, trace S1/NG signaling to check whether the dedicated bearers have been successfully set up.
- Step 6** Enable UE 1 and UE 2 to perform uplink packet injection with injection rates higher than the bandwidth of the corresponding transmission resource groups.
- Step 7** Run the **DSP RSCGRP** (old model)/**DSP IPRSCGRP** (new model) command to check whether the value of **Non-Realtime TX Bandwidth** is consistent with the transmit bandwidth configured for the transmission resource group. Traffic shaping of transmission resource groups applies only to the uplink for flow control. The bandwidth after traffic shaping should not exceed the bandwidth configured for the transmission resource group.

On the service server of the core network, check whether the traffic proportion between UEs 1 and 2 is consistent with the proportion of uplink flow control priority weight factors between UEs 1 and 2 (configured in [Step 4](#)). If the two proportions are consistent, the backpressure algorithm has taken effect.

- Step 8** Run the **SET RSCGRP** (old model)/**SET IPRSCGRP** (new model) command to turn off the traffic shaping switch and backpressure algorithm switch.
- Step 9** Enable UE 1 and UE 2 to continue uplink packet injection.
- Step 10** Run the **DSP RSCGRP** (old model)/**DSP IPRSCGRP** (new model) command to check whether the value of **Non-Realtime TX Bandwidth** is consistent with the transmit bandwidth configured for the transmission resource group. After the traffic shaping switch is turned off, the packet injection rates are not limited within the bandwidth configured for the transmission resource group. Therefore, the traffic proportion between UEs 1 and 2 (which can be obtained on the service server of the core network) is inconsistent with the proportion of uplink flow control priority weight factors between UEs 1 and 2 (configured in [Step 4](#)).
- Step 11** Run the **SET RSCGRP** (old model)/**SET IPRSCGRP** (new model) and **MOD GNBFLOWCTRLPARAMGRP** commands to restore the environment configurations.

----End

5.4.3 Network Monitoring

Transmission Port Monitoring

To create a monitoring task on the MAE-Access, perform the following steps:

- Step 1** Choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree of the displayed **Signaling Trace Management** tab page, choose **Trace Type > Base Station Device and Transport > Transport Performance Monitoring > Transport Port Traffic Monitoring**. Double-click **Transport Port Traffic Monitoring**. Select a port type for monitoring (physical port or transmission resource group). Select a monitoring type (traffic volume statistics).
- Step 2** Select different objects to be monitored and view the monitoring results as follows:
- To monitor a physical port, set **Port type** to **Physical port** and **Protocol type** to **IP**.
 - If **Physical port type** is set to **TUNNEL**, the transmit and receive rates at the network layer are measured for IP ports.
 - If **Physical port type** is set to other values, the transmit and receive rates at the data link layer are measured for IP ports.
 - To monitor a transmission resource group, set **Port type** to **RSCGRP** and view the real-time traffic volume of the transmission resource group.

----End

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *X2 and S1 Self-Management in NSA Networking*
- *NG and Xn Self-Management*
- *IPv4 Transmission*
- *Transmission Resource Management in eRAN Feature Documentation*
- *gNodeB Initial Configuration Guide (MML-based) in 3900 & 5900 Series Base Station Product Documentation*
- *X2 Self-Management in SA Networking*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*